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Integrating Electrical Resistivity Tomography (ERT) and 2D Seismic Surveys to Image a Shallow Deformation Zone

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Abstract

Near-surface deformation zones pose significant imaging challenges due to complex structural geometries, heterogeneous lithology, and high ambient noise, especially near urban and infrastructure developments. This study examines a 30-meter-wide deformation outcrop within the Dammam Dome in Eastern Saudi Arabia using electrical resistivity tomography (ERT), seismic tomography, and mineralogical analysis to characterize subsurface properties. ERT surveys revealed a pronounced high-resistivity anomaly ($\sim 2\text{--}2.4$ Ohm.m on the logarithm scale). This anomaly correlates spatially with surface fractures and is interpreted as lithified breccia or dolomitic cementation, indicative of reduced porosity and enhanced structural rigidity. In addition to the ERT data, a 2D seismic refraction survey was conducted along a parallel 360-meter profile. Super-virtual refraction interferometry (SVI) was utilized to enhance the signal-to-noise ratio and improve the continuity of first-arrival traveltimes in this noisy environment. Seismic velocity tomograms revealed a high-velocity anomaly (1210–1650 m/s) within the upper 20 meters, aligning closely with the resistivity-defined deformation zone. Mineralogical analysis via XRD and XRF confirmed dolomite as the dominant mineral, accompanied by gypsum, quartz, and clay minerals. Elevated magnesium, calcium, aluminum, and silicon concentrations further supported the interpretation of a closed-system mineralization process consistent with internally sourced diagenetic infill. These findings indicate a multi-stage deformation model involving initial faulting, subsequent brecciation, and mineral sealing. This integrated methodology of ERT, seismic tomography, and mineralogical validation provides robust insights for imaging and understanding shallow deformation zones in carbonate geological environments, significantly contributing to geohazard assessments and subsurface characterization for infrastructure development.

Introduction

Electrical resistivity tomography (ERT) and seismic methods are among the most effective geophysical techniques for shallow subsurface investigations, particularly in geologically complex environments. ERT has been widely applied in hydrogeological studies due to its sensitivity to variations in electrical resistivity caused by changes in lithology, porosity, fluid saturation, and clay content. More recently, ERT applications have expanded to structural geology, where it is used to detect and delineate faults, fractures,

and deformation zones. Resistivity variations effectively reflect contrasts in rock composition, mineral alteration, fracture-related porosity, moisture content, and compaction, especially within structurally disturbed zones (Cardarelli et al., 2010). The 2D ERT technique is particularly well-suited for mapping lateral resistivity changes by deploying multiple electrodes along linear profiles to inject electrical current and measure resulting potentials. These data are then processed through inversion algorithms to generate detailed resistivity sections that are essential for accurate geological interpretation.

Complementing ERT, seismic methods, especially seismic refraction tomography, provide valuable insights through their sensitivity to acoustic impedance contrasts resulting from variations in seismic velocity and rock density. Surface generated seismic waves propagate through subsurface layers, refracting or reflecting at geological boundaries where impedance differences occur. This allows the identification of subsurface stratigraphy, fault zones, and lithological boundaries (Hanafy, 2013). By analyzing traveltime data recorded by geophone arrays, high-resolution velocity-depth models can be constructed, offering critical information on the mechanical and lithological characteristics of subsurface materials. While ERT and seismic methods independently offer valuable insights, their integration provides a more comprehensive understanding of subsurface conditions. For example, correlating low resistivity zones from ERT with low-velocity anomalies from seismic refraction often indicates the presence of unconsolidated, water-saturated, or clay-rich sediments. Conversely, high resistivity coupled with high seismic velocities typically corresponds to compact, unfractured bedrock. When interpreted together, these complementary datasets help to distinguish lithological boundaries from fluid-induced anomalies or structural discontinuities. Moreover, combining electrical and mechanical properties allows for the detection of subtle features, such as fractured zones with partial saturation or cemented fault gouges that may be overlooked when using a single method. This joint interpretation enhances the resolution and reliability of subsurface models, particularly in geologically heterogeneous or tectonically active regions.

To further enhance geophysical interpretations, geochemical techniques such as X-ray diffraction (XRD) and X-ray fluorescence (XRF) are employed on rock samples from deformation zones. XRD identifies the dominant mineralogical phases, commonly detecting minerals such as dolomite, gypsum, quartz, and clay minerals. XRF, in turn, quantifies elemental concentrations, often revealing elevated levels of magnesium, calcium, aluminum, and silicon, signatures of dolomitization and sedimentary infill processes. These geochemical fingerprints complement geophysical findings by confirming mineralogical alternations and diagenetic processes within deformation structures, particularly in closed-system environments.

While both ERT and seismic methods offer powerful subsurface imaging capabilities, their interpretations can suffer from ambiguities due to overlapping physical responses, particularly in heterogeneous geological settings. Integrating ERT and seismic tomography with XRD and XRF analyses greatly enhances interpretive confidence. This multidisciplinary approach combines the strength of each method, enabling cross-validation and improving the resolution and reliability of subsurface models. By correlating resistivity anomalies, seismic velocity variations, and geochemical signatures, researchers can more accurately delineate structural features, assess geohazards, and develop robust geological models. Such integration is vital for infrastructure planning, resource assessment, and geological risk mitigation in complex terrains.

Geological and Structural Setting

The study area is located approximately 2.7 kilometers north of Ithra (King Abdulaziz Center of World Culture), on the eastern side of the Dharan-Jubail highway in the Eastern Province of Saudi Arabia (Figure 1a). This location lies within the broader structural feature known as the Dammam Dome, one of the most prominent hydrocarbon traps in the region. The Dammam Dome spans an area of about 500 km² and is characterized by its broad, oval-shaped structure, geographically bounded between longitudes 50°00' and 50°14' E and latitudes 26°12' and 26°26' N (Alkhalifa & Kurison, 2022). The dome includes an

extensive elliptical uplift of sedimentary rocks, particularly visible in its northern section. Among its notable geomorphological features is Jabal Um Er Rus, which, along with other structural elements, is widely believed to be the result of salt tectonics and diapiric activity (Weijermars, 1999).

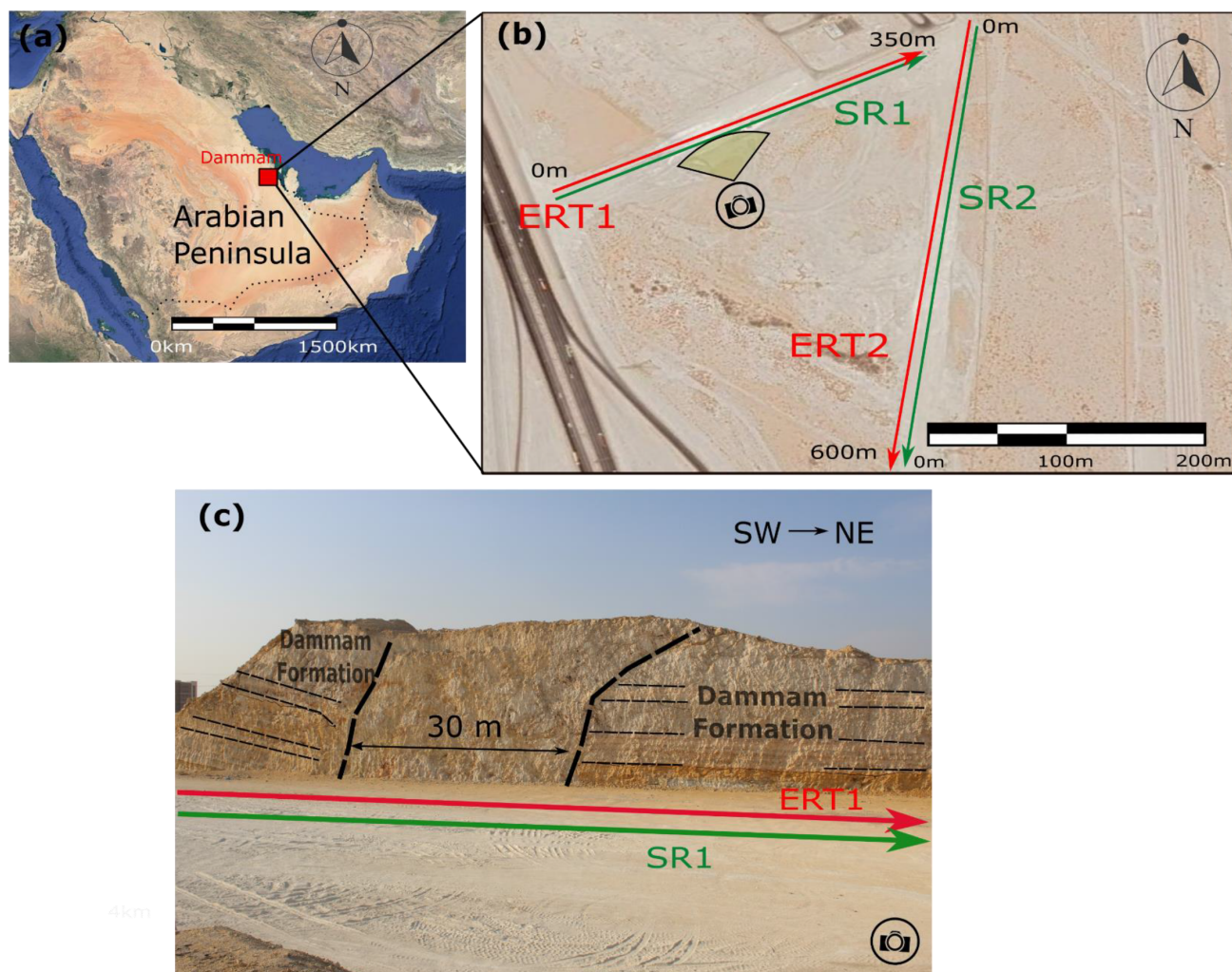


Figure 1—A) Satellite map showing the study area within the Damman Dome, Eastern Saudi Arabia. B) Aerial view with seismic (green) and resistivity (red) survey lines. C) Outcrop view of the Damman Formation along the roadcut, highlighting a 30 m wide deformation zone intersected by ERT1 and SR1.

Geological investigations in the region have identified four stratigraphic formations cropping out or underlying the area: from bottom to top, these include Rus Formation, Damman Formation, Hadruk Formation, and Dam Formation (Fig. 2). Among these, the Rus Formation is considered the regional bedrock and has been laterally traced across the area in previous studies (Weijermars, 1999). These formations collectively represent the sedimentary evolution of the Damman Peninsula and play a significant role in both hydrocarbon entrapment and groundwater dynamics.

A 500 m long road cut oriented NE-SW across the Damman Formation provides a valuable natural exposure of structural and stratigraphic analysis. The Damman Formation, which directly overlies the Rus Formation, is stratigraphically subdivided into five distinct based on lithological characteristics, originally described by Bramkamp (1941, unpublished Saudi Aramco report) and further refined by Powers et al. (1966). These members, listed from bottom to top, include: (1) Midra Shale, (2) Saila Shale, (3) Alveolina Limestone, (4) Khobar Limestone, and (5) Alat Dolomitic Limestone/Marl (Fig. 2). This stratigraphic sequence is widely exposed across the Damman Dome and serves as a reference framework for structural

and hydrogeological interpretations. Recent work by [Alkhalifa & Kurison \(2022\)](#) interpreted a prominent slope on the northwestern side of the outcrop as a collapse zone, approximately 28 m wide along a NNW-SSE orientation ([Fig. 1c](#)). This feature suggests post-depositional deformation potentially linked to subsurface dissolution or fault displacement, which aligns with regional tectonic models.

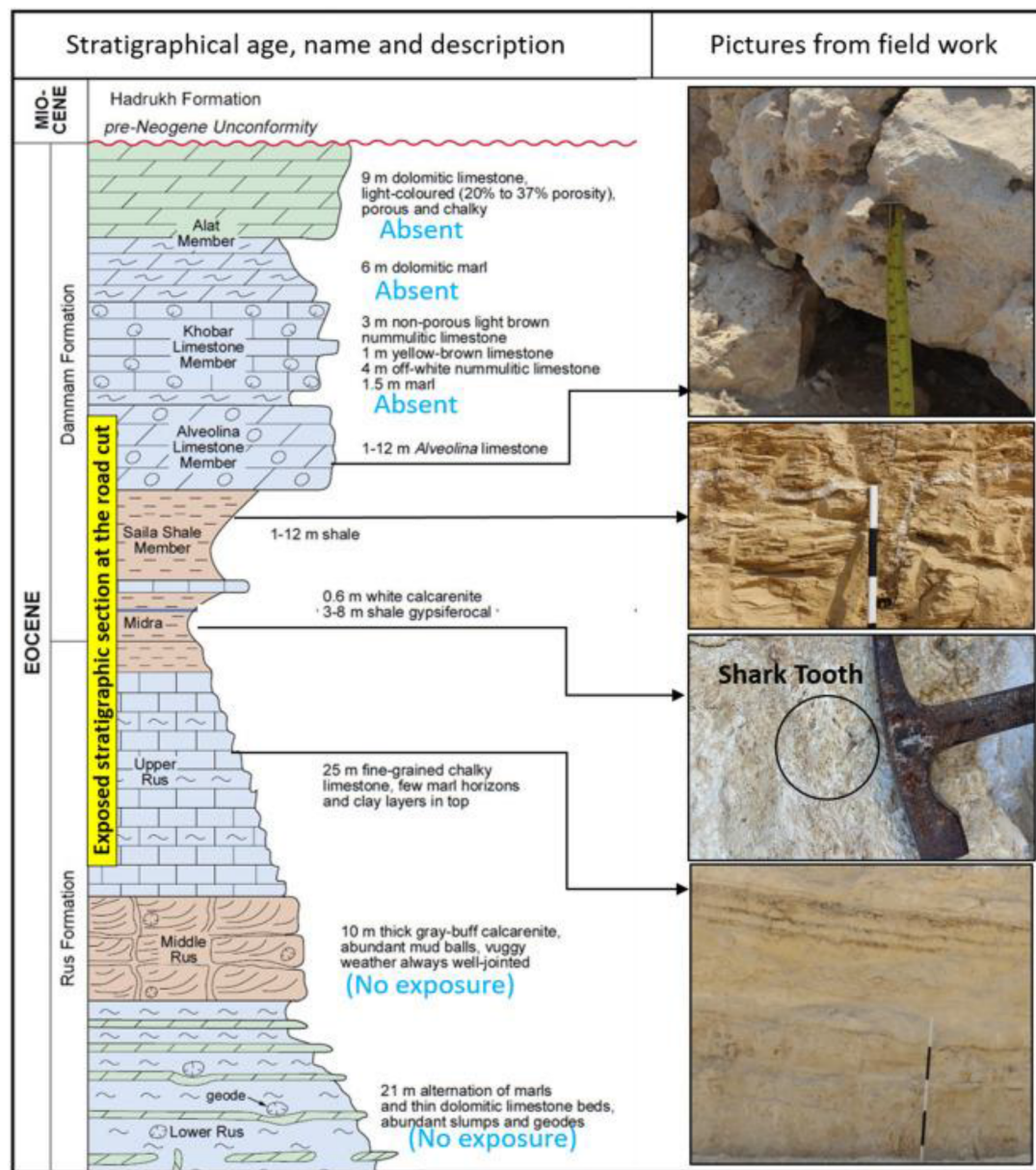


Figure 2—The stratigraphic column describing different formations in the area (after [Weijermars, 1999](#); [Powers, 1966](#); [Tleel, 1973](#)).

The broader Dammam Peninsula includes three major formations observed at the surface: the Dam Formation (Td), the Dammam Formation (Tdm), and the Upper Rus Formation (Tr"u"). Among these, the Dammam formation is the dominant unit exposed in the study area ([Fig. 2](#)) and serves as the main subject of structural and stratigraphic investigation. The outcrop under study offers a rare and well-preserved opportunity to observe and document a deformation zone, with extensive evidence of faulting and veining system, likely associated with tectonic activity and diagenetic processes. Field observations along the newly cut road section revealed a network of natural fractures, including normal faults, shear fractures, and

gypsum-filled veins, particularly concentrated within the deformation zone but also extending into adjacent layers of the Dammam formation (Fig. 1c).

Methods

The seismic refraction profile (SR) was acquired using a linear array of 120 single-component vertical geophones, spaced at 3 m intervals, Resulting in a total profile length of 360 m oriented in the SW-NE direction. A total of 120 shot gathers were recorded, with one shot placed at each geophone location, maintaining a consistent 3 m shot interval. This configuration ensured uniform energy input and optimized subsurface coverage across the entire profile. Data acquisition was carried out using a Geode seismic system, known for its robustness and precision in shallow seismic surveys. Seismic energy was generated using a 200-lb accelerated weight drop system, a reliable source for high-resolution near surface imaging (see Table 1 for acquisition parameters).

Table 1—Parameters for the 2D seismic survey.

Source	200-lb weight drop
Recording system	Geode
No. of shots	120
No. of receivers	120
Shot spacing	3 m
Receiver spacing	3 m
Survey length	360 m
No. of tracers	14,400
Sampling interval	0.2 ms
Record length	1.0 s

The initial ERT profile was recorded along a SW-NE trending profile, aligned with the previously recorded seismic line to enable direct geophysical correlation. The data was recorded using a Syscal Pro Switch 96 multi-electrode resistivity system from IRIS Instruments. The array consisted of 96 electrodes spaced 5 m intervals, covering a total profile length of 480 m. A dipole-dipole (DD) configuration was deployed due to its high sensitivity to horizontal resistivity contrasts and its effectiveness in detecting vertical and sub-vertical structural, including faults, fractures, and deformation zones. As noted by Reynolds (2011), the DD array is particularly well-suited for imaging lateral resistivity variations, which are characteristic of structurally complex zones in sedimentary basins.

To ensure data integrity, quality control procedures were implemented to identify and remove unreliable resistivity measurements, such as anomalously high or noisy data points (spikes). This step is critical for improving the quality of the raw ERT dataset and minimizing inversion errors in subsequent processing. In parallel, first break picking was performed for the seismic dataset to prepare for refraction analysis. This accomplished using an in-house software tool called DataPick, which allows for efficient and accurate identification of first arrival times. Depending on the signal-to-noise ratio (SNR) of the recorded seismic traces, the picking process can be labor-intensive, especially in areas with weak or dispersed energy arrivals. Figure 3 shows an example of shots gathered with the first arrivals picked. Following the initial data preparation, Super-Virtual Interferometry(SVI) was applied to derive a velocity model of the subsurface. To address uneven ray coverage and enhance image continuity, a rectangular smoothing filter was applied duringr each iteration of the inversion process following the methodology of Nemeth et al. (1997).This smoothing technique reduces the influence of localized anomalies or gaps in ray density that can introduce artifacts in the velocity model. The smoothing grid began with a size of 20 m × 10 m and was progressively

refined—halved after the fifth iteration, until the desired resolution was achieved. This iterative approach helped balance model sharpness with stability, ultimately producing a cleaner and more interpretable seismic image. The smoothing parameters used during inversion are summarized in Table 2.

Table 2—Smoothing schedule of 2D data with 5 iterations each.

Grid size	0.5 m
Smoothing size 1	20 × 10 (pixels)
Smoothing size 2	12 × 6 (pixels)
Smoothing size 3	8 × 4 (pixels)
Smoothing size 4	4 × 2 (pixels)

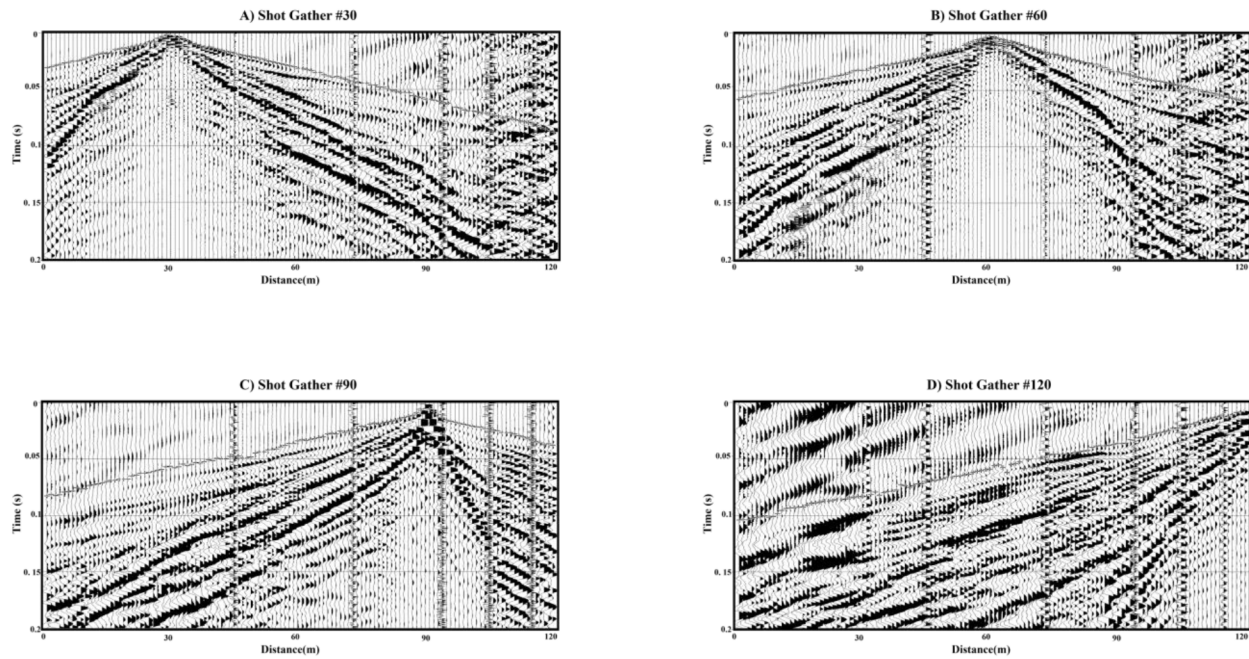


Figure 3—Samples of the recorded shot gathers from different locations to show the quality of the recorded data. The displayed shot gathers are a) 1, b) 60, c) 90, and d) 120.

The SVI method is a technique used in seismic data processing to enhance the signal-to-noise ratio (SNR) of refraction arrivals. The method was originally described by Bharadwaj and Schuster (2010) and Bharadwaj and Schuster (2012), as well as by Bharadwaj et al. (2011) and Mallinson et al. (2011). SVI method employs the far-field reciprocity equation of both correlation (Eq. 1) and convolution (Eq. 2) types to create super-virtual refractions. This results in an enhanced SNR by a factor of $\sqrt{N}$, where N is the number of source positions associated with the generation of the head-wave arrival and coincident with the receiver.

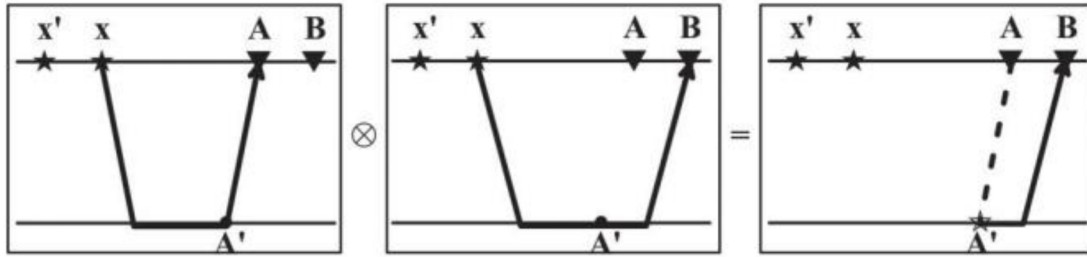
$$\text{Im}[G(A|B)^{\text{virt}}] \approx \sim k \int G(A|X) * G(B|X) dx \quad (1)$$

$$G(B|X')^{\text{super}} \approx 2ik \int G(A|B)^{\text{virt}} \cdot G(A|X') dA \quad (2)$$

In the SVI method, the source positions are denoted by x in Figure 5(a) and x' in Figure 5(b), while the receivers are located at A and B . The average wavenumber is represented by k . $G(A|B)$ denotes the head-wave contribution in Green's function for a virtual shot at A' , a receiver at B , and the virtual excitation time is equal to the negative traveltimes from A to A' . On the other hand, $G(B|x')^{\text{super}}$ represents the super-virtual data obtained by convolving the recorded data $G(A|x')$ with the virtual data $G(A|B)^{\text{virt}}$, as shown in Fig.

4b). This convolution process enhances the SNR of the refraction arrivals by combining the recorded data with virtual data generated from the head-wave contribution in Green's function.

a) Crosscorrelate and stack over x to generate virtual refractions



b) Convolve and stack over A to generate super-virtual refractions

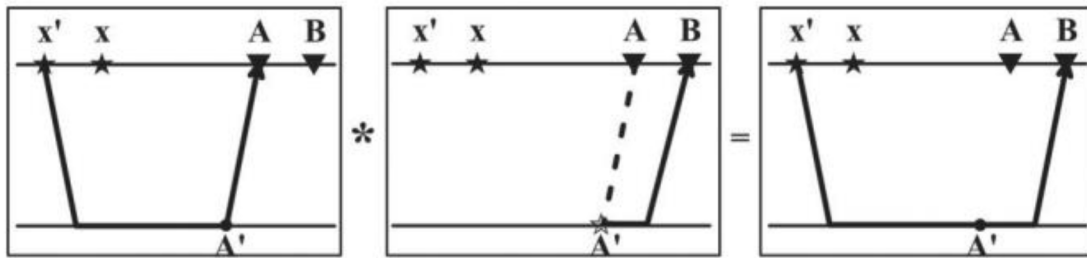


Figure 4—The steps for creating super-virtual refraction arrivals. a) Correlation of the recorded trace at A with that at B for a source at x to give the trace $G(A,B,t)$ with the virtual refraction having traveltime denoted by $\tau(A'B) - \tau A'A$. This arrival time will be the same for all post-critical source positions, so stacking $\Sigma G(A,B,t)$ will enhance the SNR of the virtual refraction by $\sqrt{N}$. b) Similar to that in a) except the virtual refraction traces are convolved with the actual refraction traces and stacked for different geophone positions to give the super-virtual trace with a SNR enhanced by $\sqrt{N}$. Here, N denotes the number of coincident source and receiver positions that are at post-critical offset for this particular refraction (after Bharadwaj).

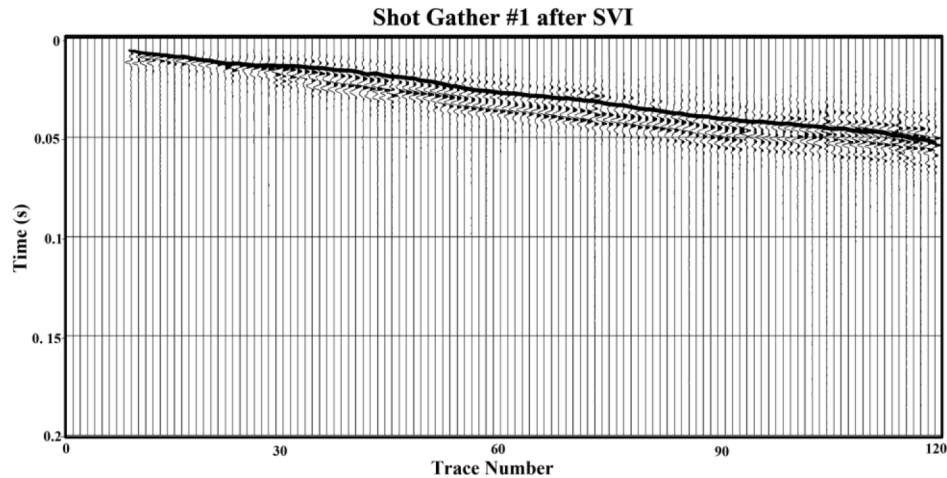


Figure 5—Shot gather #1 from the recorded 2D seismic data. The solid black line is the first break picking after applying SVI. Traces show an increase in signal-to-noise ratio, which enables picking of all recorded traces in this shot gather.

As mentioned earlier, the SVI method may produce artifacts in the super-virtual data set due to the limited recording aperture and discrete sampling. To avoid these artifacts, Dong et al. (2006) suggested a windowing technique. The workflow of the SVI method with windowing is as follows:

- Filter the raw data to remove high-frequency noise.
- Window about the first arrivals. The suggested window length is 1 period before the expected first-arrival times and 2 to 3 periods after it. This windowing technique ensures that only head-wave arrivals, which are less affected by artifacts, are correlated with one another.
- Use Eq. (1) to generate virtual traces.
- Use Eq. (2) to generate super-virtual traces.

Results

The analysis of the 2D seismic velocity tomogram has provided valuable insights into the subsurface structure of the study area. Within the upper 20 meters of the profile, a distinct zone of high-velocity values was identified, spatially coinciding with the surface expression of the deformation zone. This correlation strongly suggests that the observed high-velocity anomalies are linked to structural features such as compacted or cemented fault zones, fracture healing, or mineralized veins, all of which can significantly increase the seismic wave propagation speed. The velocity values within this zone range from 1210 m/s to 1665 m/s, clearer higher than those in surrounding areas. These elevated velocities may reflect increased rock stiffness and density due to diagenetic cementation, reduced porosity, or stress-induced compaction, typically associated with long-lived or active deformation systems.

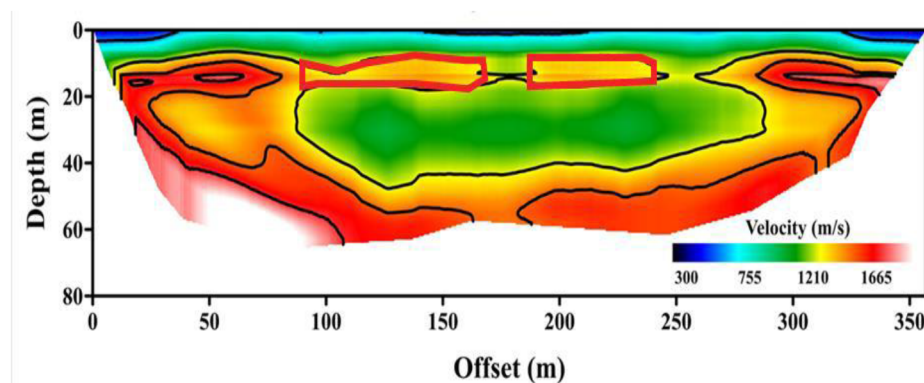


Figure 6—The first arrival traveltimes tomogram shows a highlighted area in the top 20 meters of the subsurface, where high-velocity anomalies are present and coincide with the location of the deformation zone. The velocity values of these anomalies range between 1210 and 1665 m/s.

The 2D ERT survey revealed a prominent high-resistivity anomaly that correlates spatially with a similar feature identified in the 2D seismic velocity tomogram. This spatial correspondence between two independent geophysical datasets enhances the confidence and credibility of the interpreted subsurface features, affirming the robustness of the integrated approach. The resistive anomaly, ranging from 2.0 to 2.4 Ohm.m on a logarithm scale, marks a distinct contrast in the subsurface's electrical properties. Such elevated resistivity values are often indicative of densely compacted, low-porosity materials, cemented fracture zones, or dry or resistive lithologies, potentially associated with deformation-related processes such as mineral precipitation or structural compaction. When combined with the high seismic velocities observed in the same region, the anomaly suggests a zone of increased stiffness and reduced fluid content, likely tied to a well-developed fault core or deformation band.

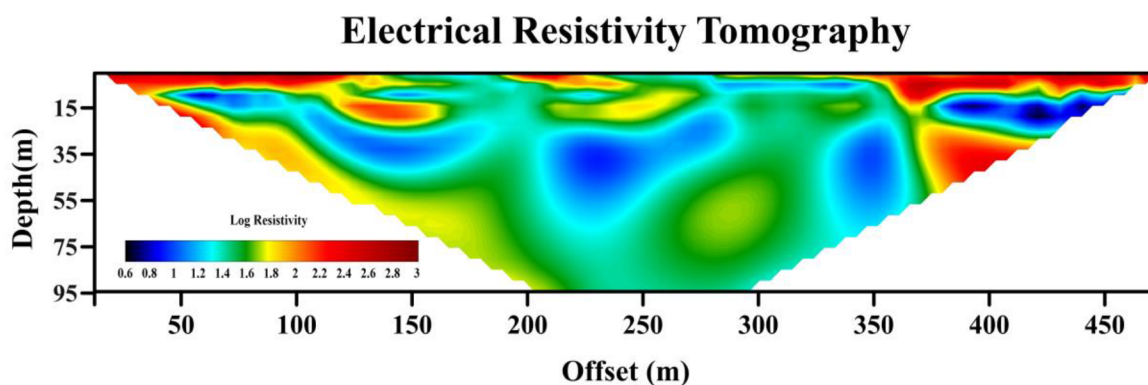


Figure 7—Electrical Resistivity Tomogram shows a maximum depth of 95 m and a high resistivity value of 3 Ohm/m on a logarithmic scale.

Mineralogical compositions of the host lithologies and fracture cement were measured for representative samples collected from the study area to characterize the fracture geochemistry within the deformation zone (Figure 8).



Figure 8—Samples 1&2 collected from fracture filling of the right and left wall of the deformed zone. Where sample 3 was collected from the host rock within the deformed zone close to the small fault. Sample 4 was taken from a small fracture within the deformation zone. Sample 5 was collected from the bottom of the deformation zone. The 6th sample was from the host rock on the bottom right outside of the deformation zone.

Samples 1 and 5 were extracted from the left boundary/wall of the deformation zone with host rocks, sample 6 was extracted from the right boundary of the deformation zone with host rocks and samples 2, 3, and 4 were extracted from the deformed interval and fracture/fault cement within the deformation zone. The XRD and XRF result of the six samples (Figures 9 & 10) show that the filling material of samples 1, 2 and 4 are gypsum, quartz, dolomite, and clay minerals. Moreover, the dominant mineral in the host rock within the deformation zone (samples 3 & 5) is dolomite. The results of sample 6 show the dominance of clay minerals.

The mineralogical analysis of the six samples reveals a diverse range of lithologies and depositional environments. Sample 1 is dominated by dolomite (~72%) with minor quartz (~22%) and trace amounts of clay minerals (~3%), suggesting a carbonate-rich composition typical of dolostone with limited siliciclastic input. In contrast, Sample 2 is almost entirely composed of calcite (~95%), indicative of a highly pure limestone, likely formed in a stable marine environment with minimal detrital influence. Sample 3 presents a more heterogeneous composition, with significant proportions of dolomite (~36%), clay minerals (~35%), and quartz (~25%), alongside minor calcite (~0%). This mixture suggests a marl or dolomitic shale, reflecting a transitional setting between carbonate and clastic deposition. Sample 4 is unique in its mineralogy, exhibiting a high concentration of evaporites (~75%), with the remaining portion consisting of small but balanced amounts of quartz (~10%), dolomite (~5%), clay (~0%), and feldspar (~5%). This composition points to an evaporitic environment, such as a restricted marine basin or sabkha, where high evaporation rates led to significant salt precipitation. Sample 5, like Sample 1, is also dominated by dolomite

(~94%) but is even purer, indicating a massive dolostone unit with minimal clastic contamination. Lastly, Sample 6 is overwhelmingly composed of clay minerals (~78%) with a notable feldspar content (~22%), reflecting a fine-grained siliciclastic sediment, possibly a claystone or mudstone, derived from a feldspar-rich source such as granitic terrains.

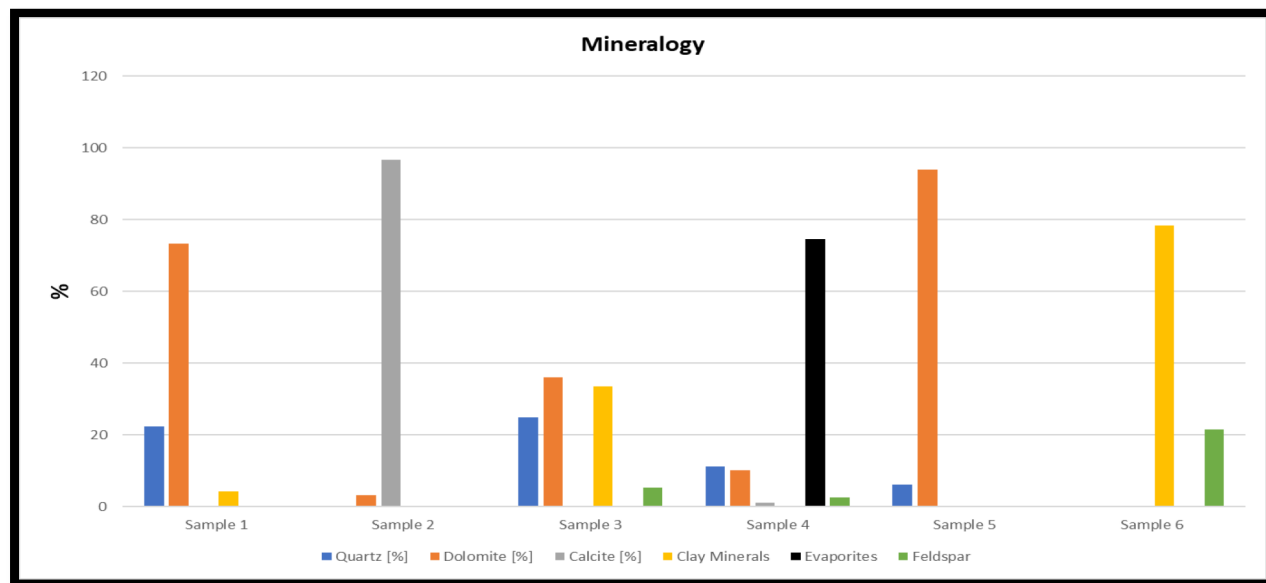


Figure 9—The analyzed samples 1, 2, and 4 exhibit filling materials composed of gypsum, quartz, dolomite, quartz, and clay minerals. Furthermore, samples 3 and 5, taken from the deformation zone, indicate that dolomite is the dominant mineral in the host rock. In contrast, sample 6 is characterized by the prevalence of clay minerals.

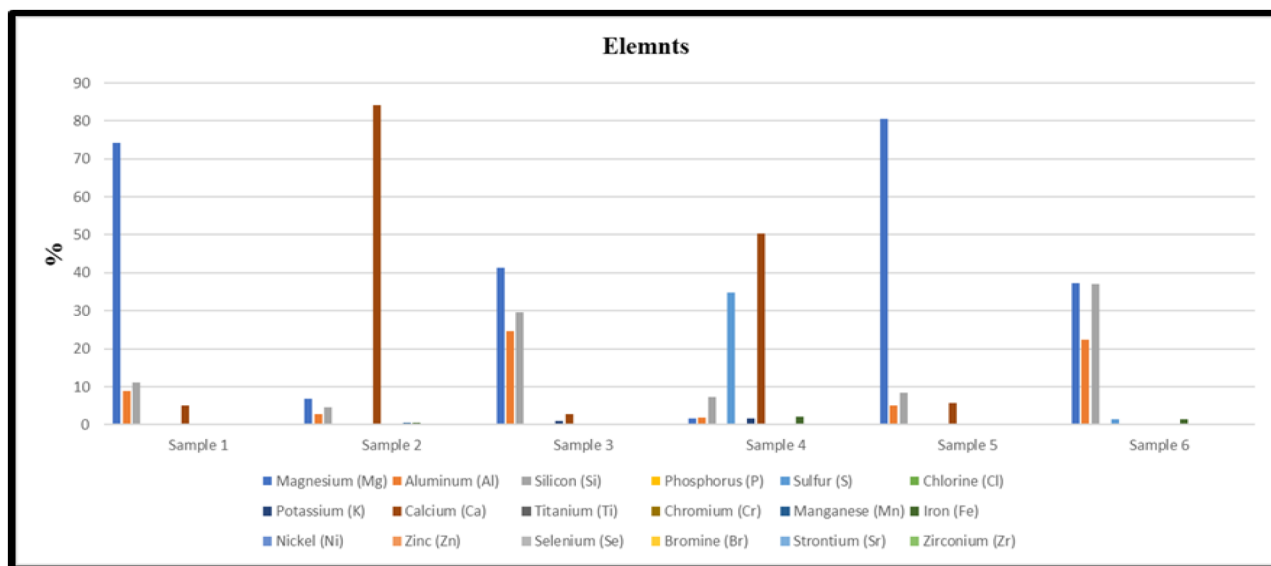


Figure 10—The XRF result agrees with the finding of XRD, where magnesium and calcium are the most common elements in most samples. Furthermore, sample #6 shows that the filling material has elements (Mg, Al, Si).

The elemental analysis highlights clear compositional differences across the six samples. Sample 1 and Sample 5 show high magnesium (Mg) content (~75%), consistent with dolomite-rich lithologies. Sample 2 is dominated by calcium (Ca) (~86%), confirming a pure limestone composition. Sample 3 has a balanced mix of Mg, Ca, Al, and Si, reflecting a marl or dolomitic shale with mixed carbonate and siliciclastic input. Sample 4 contains high sulfur (S) and chlorine (Cl), indicating evaporite minerals like gypsum or halite, alongside elevated Mg, suggesting dolomitic influence. Sample 6 is enriched in aluminum (Al), silicon

(Si), and potassium (K), pointing to a clay- rich, fine-grained sediment with feldspar and aluminosilicate components.

The deformation zone evolved from one significant fault, which evolved into a fault zone due to the minor faults we saw during the geological investigation (Figure 12). Later, this fault zone evolved to become a shear zone in the final stage. Figure 11 shows the formation stages of a shear zone, which is the interpretation of this study's deformation zone formation stages. This is evident by the mineralogical composition results of the fracture cement samples, indicating that the filling materials of the faults observed within the deformation zone (shear zone) are derived from the same sedimentary unit surrounding the deformation zone. This suggests that the system is closed, indicating that the material within the system is not being exchanged with material from other units.

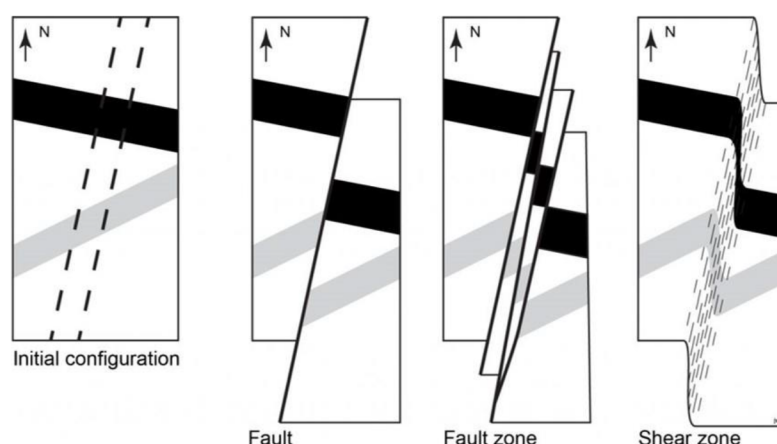


Figure 11—This schematic drawing illustrates the states believed to have resulted in the deformation zone in the study area, which is interpreted as four main sequential stages forming a shear zone.

Conclusions

The integrated application of electrical resistivity tomography (ERT), seismic refraction tomography, and mineralogical analyses has proven effective in imaging and characterizing a shallow deformation zone within the Dammam Dome. The ERT results revealed a prominent high-resistivity anomaly ($\sim 2\text{--}2.4\text{ Ohm}\cdot\text{m}$ in logarithmic scale), spatially coinciding with a mapped deformation feature. This resistive signature likely reflects lithified breccia, mineral-filled fractures, or dolomitic cementation, all indicative of reduced porosity and increased rock stiffness. Complementing these findings, 2D seismic velocity tomography identified a high-velocity anomaly ($1210\text{--}1665\text{ m/s}$) within the same near-surface interval. The velocity enhancement, supported by the application of super-virtual refraction interferometry (SVI), aligns with the resistivity data and reinforces the interpretation of a mechanically rigid and compositionally altered deformation zone. Mineralogical characterization using XRD and XRF techniques confirmed the dominance of dolomite and elevated concentrations of magnesium and calcium within the fractured intervals. These results suggest that the fracture fillings originated from surrounding sedimentary units, supporting a closed-system diagenetic model involving internal mineral mobilization and fault sealing. Overall, the study demonstrates the value of a multidisciplinary workflow that combines geophysical and geochemical datasets to identify and interpret shallow structural features. This approach enhances confidence in subsurface imaging and is highly applicable to geohazard assessment, civil planning, and resource management in carbonate terrains.

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